CloudNine Systems

Group 4 | Milestone 1 ERD

50 tables organized into seven connected modules.

Logical relational design in Third Normal Form (3NF).

Each module page shows all owned attributes and incoming FK relationships.

Gray entities are references to tables fully defined on another module page.

Relationship ends: 1 = required parent; 0..1 = optional parent;

0..* = zero or more children; 0..1 child = unique foreign key.

PK = primary key; FK = foreign key; UK = additional unique key.

? means nullable. Composite PKs are marked on every participating column.

Minimum child counts are enforced by lifecycle rules, not by ordinary FKs.

XOR, time interval, capacity and accounting rules appear in the analysis report.

All monetary values are USD. All timestamps are UTC.

The editable Mermaid and JSON sources accompany this PDF.

2. Clients and people

3. Infrastructure and history

4. Reservations and agreements

5. Deployment and consumption

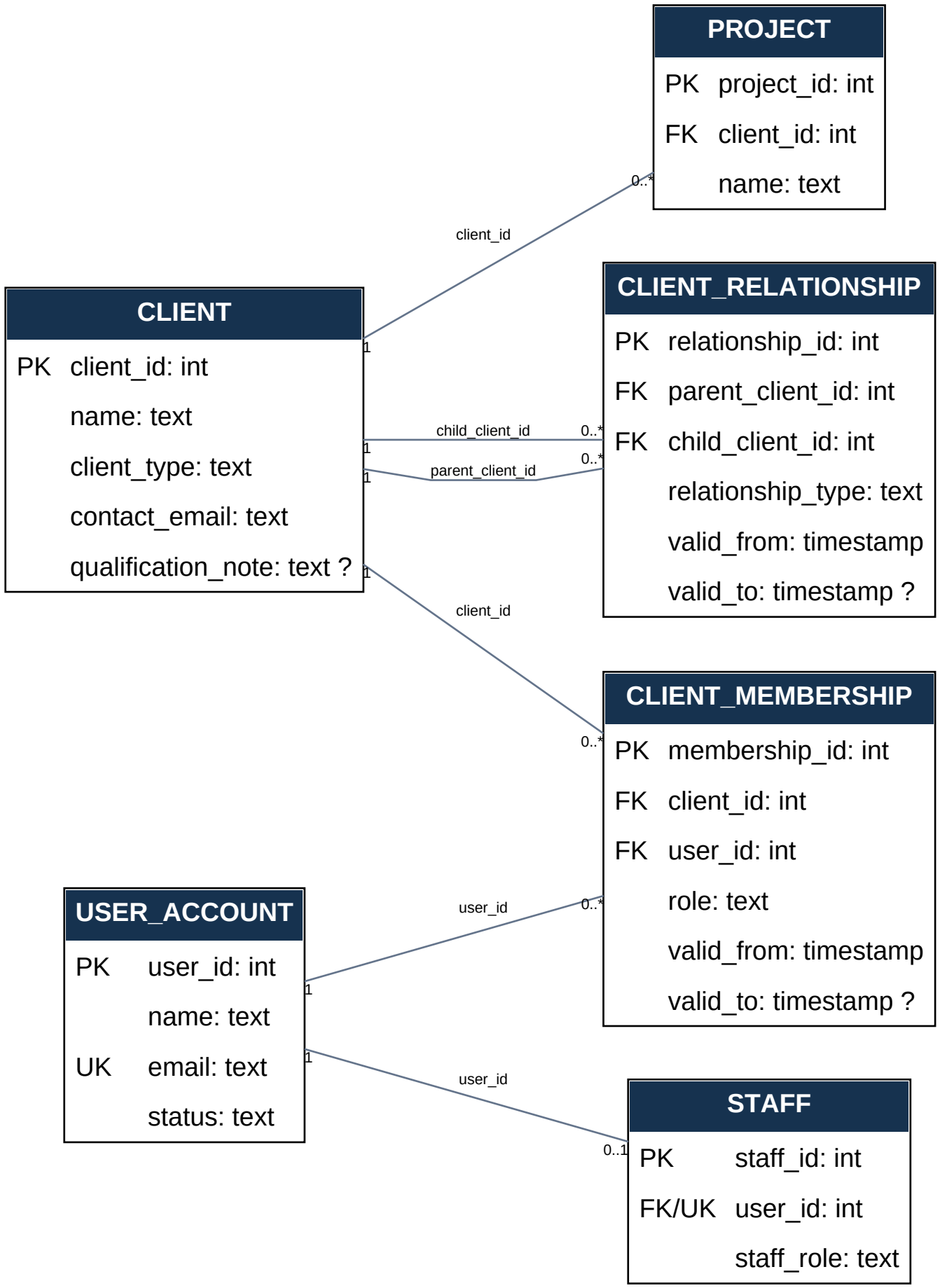
6. Pricing and charge allocation

7. Invoices and payments

8. Access and audit

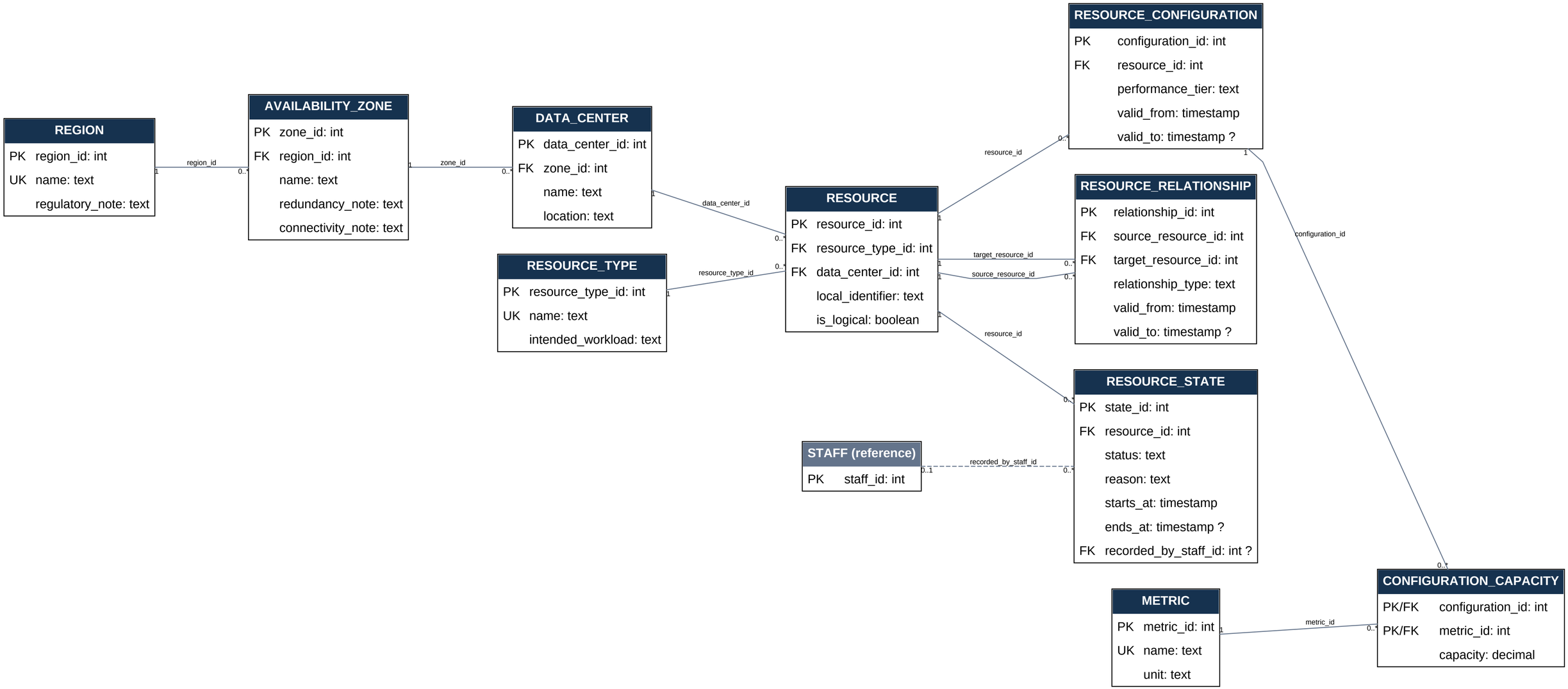
1. Clients and people

PK / FK / UK | ? nullable | Gray = external reference | Edge labels give multiplicity at each end



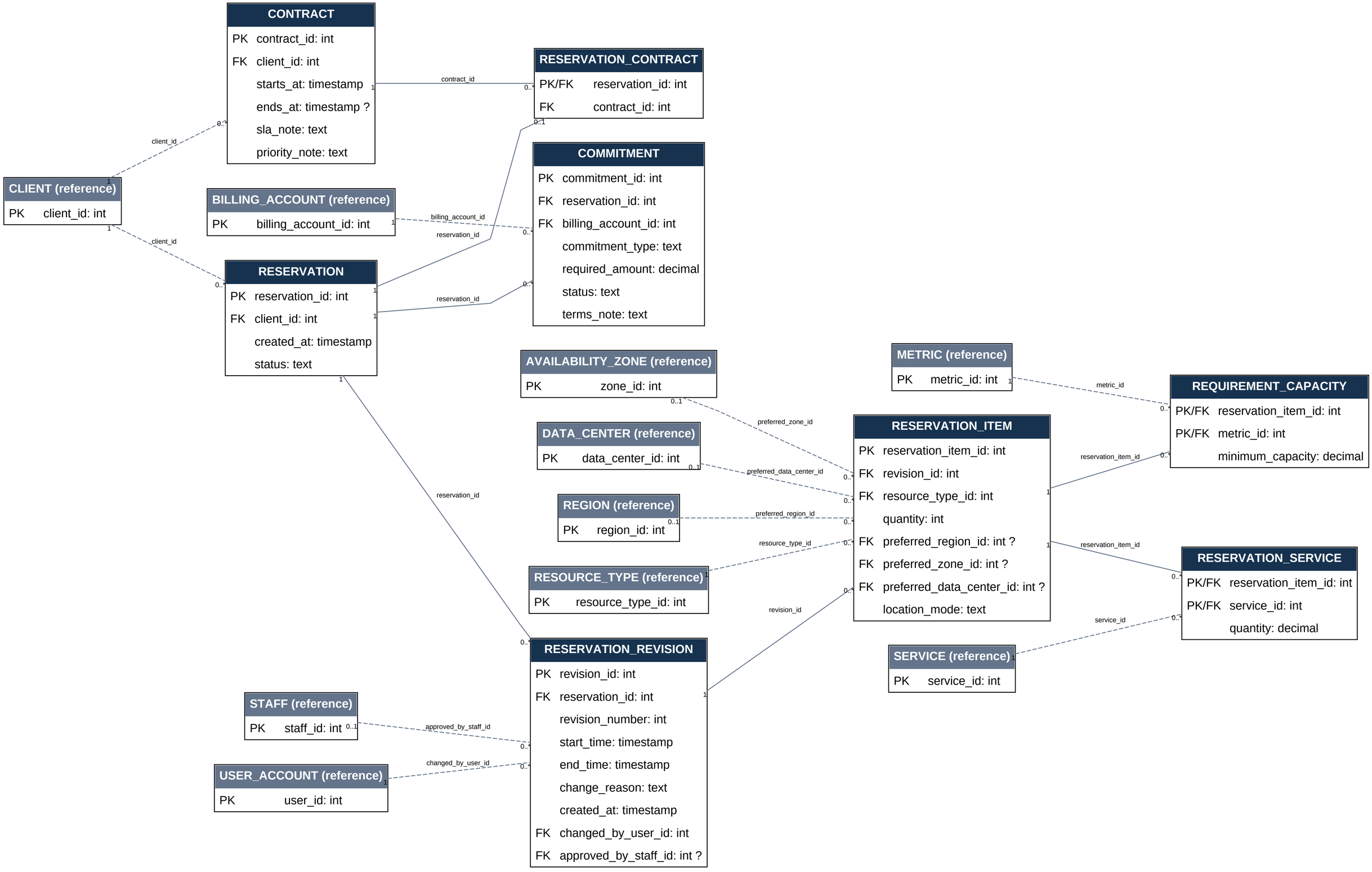
2. Infrastructure and history

PK / FK / UK | ? nullable | Gray = external reference | Edge labels give multiplicity at each end



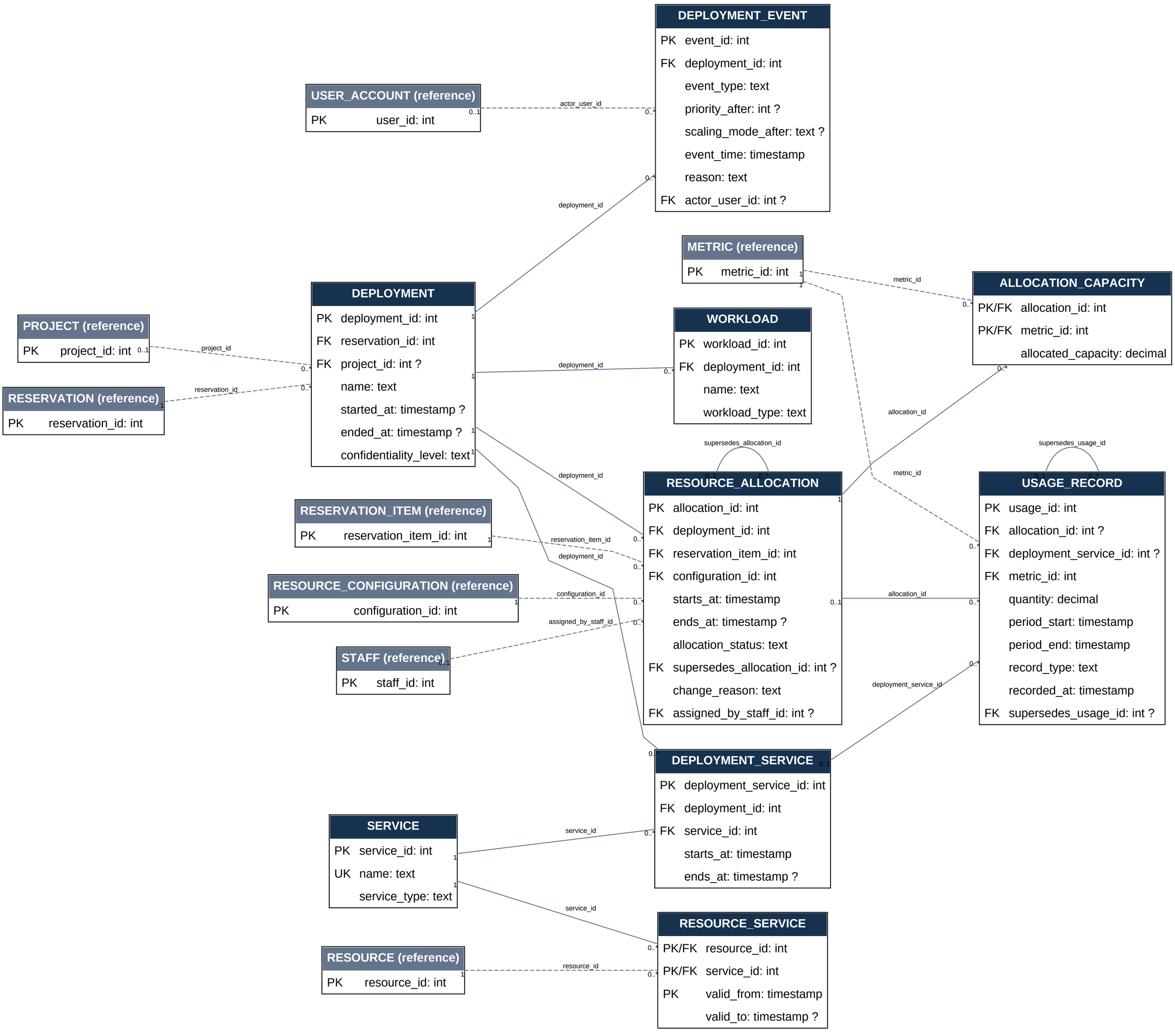
3. Reservations and agreements

PK / FK / UK | ? nullable | Gray = external reference | Edge labels give multiplicity at each end



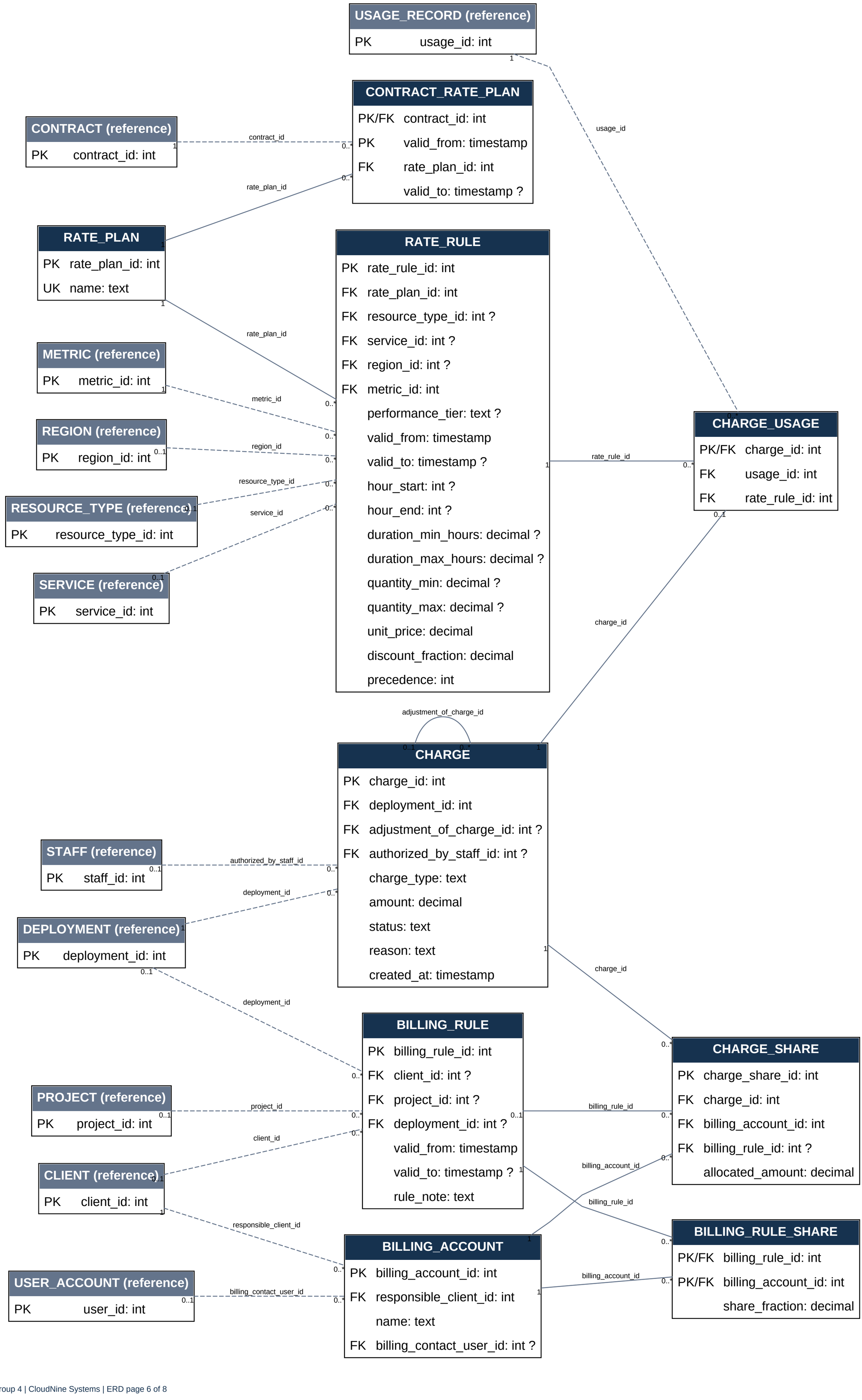
4. Deployment and consumption

PK / FK / UK | ? nullable | Gray = external reference | Edge labels give multiplicity at each end



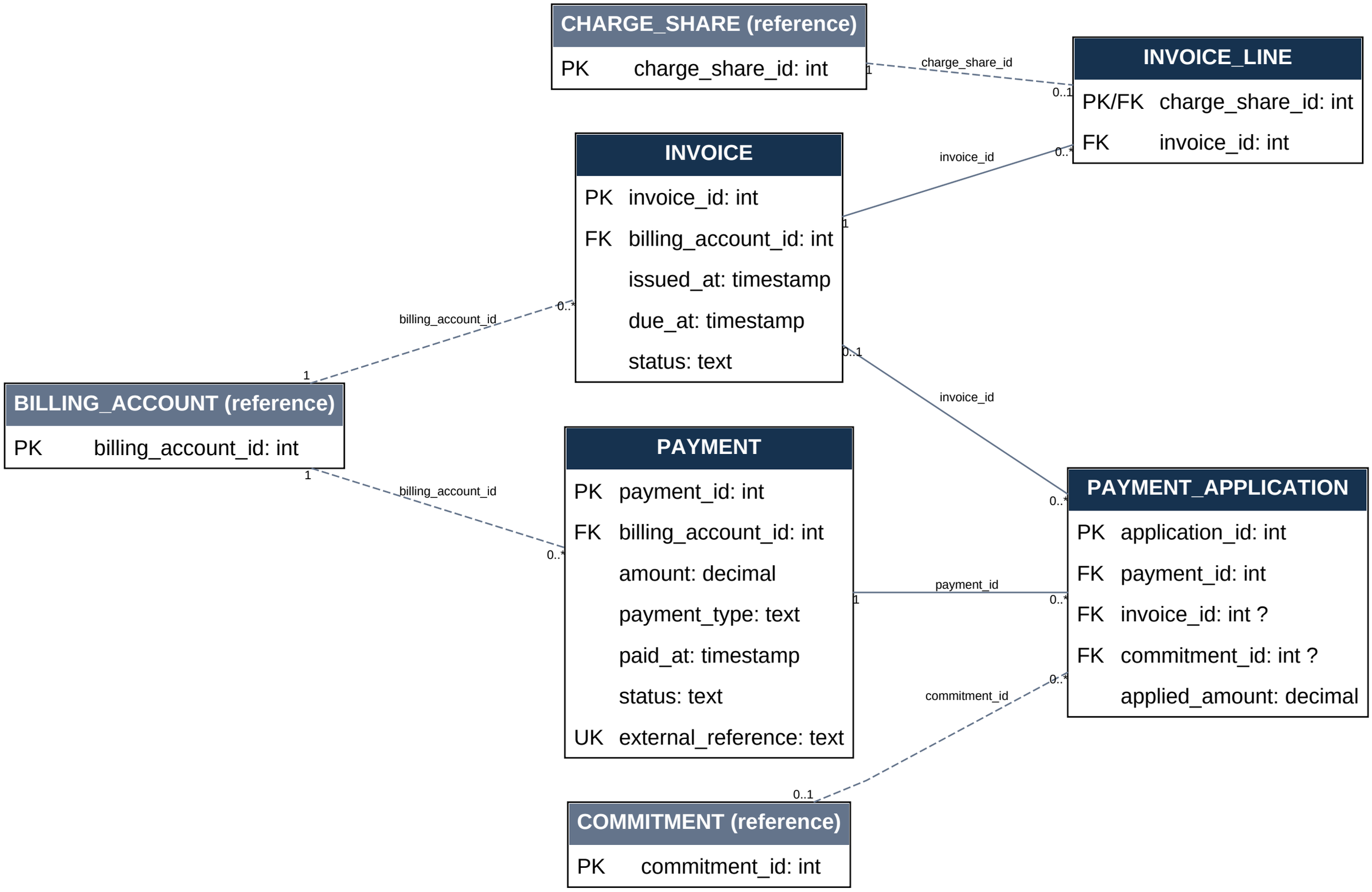
5. Pricing and charge allocation

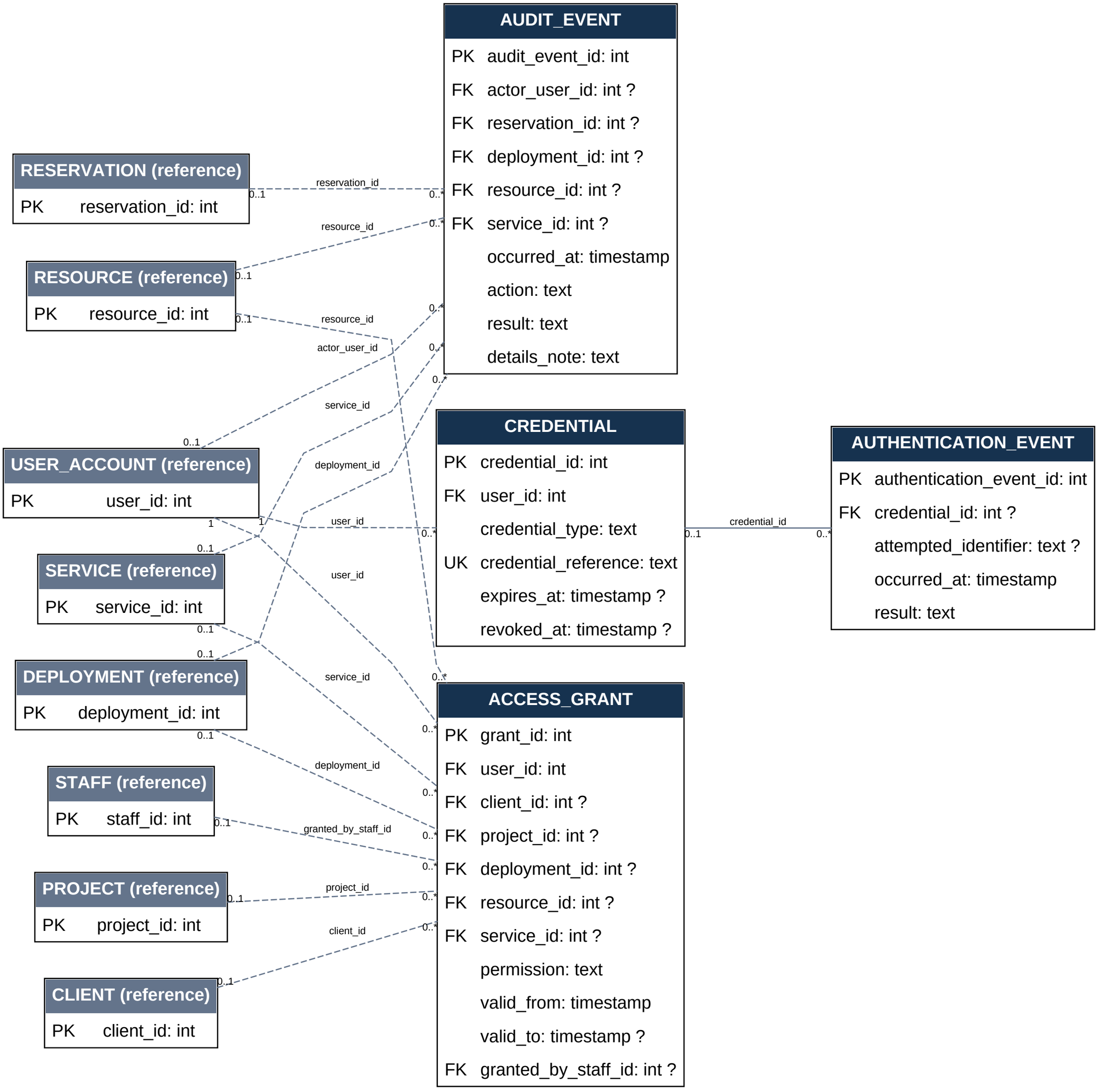
PK / FK / UK | ? nullable | Gray = external reference | Edge labels give multiplicity at each end



6. Invoices and payments

PK / FK / UK | ? nullable | Gray = external reference | Edge labels give multiplicity at each end



7. Access and audit

PK / FK / UK | ? nullable | Gray = external reference | Edge labels give multiplicity at each end